

Association of an estrogen-sensitive *Pax1-Col11a1-Mmp3* signaling axis with adolescent idiopathic scoliosis

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Hao Yu, Anas M. Khanshour, Aki Ushiki, Nao Otomo, Yoshinao Koike, Elisabet Einarsdottir, Yanhui Fan, Lilian Antunes, Yared H. Kidane, Reuel Cornelia, Rory Sheng, Yichi Zhang, Jimin Pei, Nick V. Grishin, Bret M. Evers, Jason Pui Yin Cheung, John A. Herring, Chikashi Terao, You-Qiang Song, Christina A. Gurnett, Paul Gerdhem, Shiro Ikegawa, Jonathan J. Rios, Nadav Ahituv, Carol A. Wise 

Center for Pediatric Bone Biology and Translational Research, Scottish Rite for Children, Dallas, TX, USA • Department of Bioengineering and Therapeutic Sciences, University of California San Francisco, San Francisco, CA, USA • Institute for Human Genetics, University of California San Francisco, San Francisco, CA, USA • Laboratory of Bone and Joint Diseases, RIKEN Center for Integrative Medical Sciences, Tokyo, JP • Laboratory for Statistical and Translational Genetics, RIKEN Center for Integrative Medical Sciences, Yokohama, JP • Science for Life Laboratory, Department of Gene Technology, KTH-Royal Institute of Technology, Solna, SE • School of Biomedical Sciences, The University of Hong Kong, Hong Kong SAR, CN • Department of Neurology, Washington University in St. Louis, St. Louis, MO, USA • School of Pharmaceutical Sciences, Tsinghua University, Beijing, CN • Department of Biophysics, University of Texas Southwestern Medical Center, Dallas, TX, USA • Department of Pathology, University of Texas Southwestern Medical Center, Dallas, TX, USA • Department of Ophthalmology, University of Texas Southwestern Medical Center, Dallas, TX, USA • Department of Orthopaedics and Traumatology LKS Faculty of Medicine, The University of Hong Kong, Hong Kong SAR, CN • Department of Orthopedic Surgery, Scottish Rite for Children, Dallas, TX, USA • Department of Orthopaedic Surgery, University of Texas Southwestern Medical Center, Dallas, TX, USA • Department of Clinical Science, Intervention & Technology (CLINTEC), Karolinska Institutet, Stockholm, Uppsala University, Uppsala, SE • Department of Surgical Sciences, Uppsala University and • Department of Orthopaedics and Hand Surgery, Uppsala University Hospital, Uppsala, SE • Eugene McDermott Center for Human Growth and Development, University of Texas Southwestern Medical Center, Dallas, TX, USA • Department of Pediatrics, University of Texas Southwestern Medical Center, Dallas, TX, USA

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Abstract

Adolescent idiopathic scoliosis (AIS) is a common and progressive spinal deformity in children that exhibits striking sexual dimorphism, with girls at more than five-fold greater risk of severe disease compared to boys. Despite its medical impact, the molecular mechanisms that drive AIS are largely unknown. We previously defined a female-specific AIS genetic risk locus in an enhancer near the *PAX1* gene. Here we sought to define the roles of *PAX1* and newly-identified AIS-associated genes in the developmental mechanism of AIS. In a genetic study of 9,161 individuals with AIS and 80,731 unaffected controls, significant association was identified with a variant in *COL11A1* encoding collagen (α1) XI (rs3753841; NM_080629_c.4004C>T; p.(Pro1335Leu); $P=7.07 \times 10^{-11}$, OR=1.118). Using CRISPR mutagenesis we generated *Pax1* knockout mice (*Pax1*^{-/-}). In postnatal spines we found that *Pax1* and collagen (α1) XI protein both localize within the intervertebral disc (IVD)-vertebral junction region encompassing the growth plate, with less collagen (α1) XI detected in *Pax1*^{-/-} spines compared to wildtype. By genetic targeting we found that wildtype *Col11a1* expression in growth plate cells (GPCs) suppresses expression of *Pax1* and of *Mmp3*, encoding the matrix

metalloproteinase 3 enzyme implicated in matrix remodeling. However, this suppression was abrogated in the presence of the AIS-associated *COL11A1*^{P1335L} mutant. Further, we found that either knockdown of the estrogen receptor gene *Esr2*, or tamoxifen treatment, significantly altered *Col11a1* and *Mmp3* expression in GPCs. These studies support a new molecular model of AIS pathogenesis wherein genetic variation and estrogen signaling increase disease susceptibility by altering a *Pax1-Col11a1-Mmp3* signaling axis in the growth plate.

eLife assessment

This **valuable** study analyzed a large cohort of Adolescent Idiopathic Scoliosis (AIS) patients and identified an association with a variant in *COL11A1* (Pro1335Leu). Experimental testing of this potentially pathogenic variant suggests a link between Pax1, Col11a1, Mmp3, and estrogen signaling. While the connection between transcriptional regulation, the extracellular matrix, and estrogen signaling is **compelling**, the experimental evidence supporting this link is still **incomplete** and would benefit from a more direct and comprehensive approach. Strengthening the functional testing would help establish the pathogenicity of a collagen variant and the role of the proposed Pax1-Col11a1-Mmp3 signaling axis in AIS.

Introduction

The human spinal column is a dynamic, segmented, bony and cartilaginous structure that is essential for integrating the brain and nervous system with the axial skeleton while simultaneously providing flexibility in three dimensions¹. Idiopathic scoliosis is the most common developmental disorder of the spine, typically appearing during the adolescent growth spurt. Adolescent idiopathic scoliosis (AIS) is reported in all major ancestral groups, with a population prevalence of 1.5-3%^{2,3}. Children with AIS usually present with a characteristic right-thoracic major curve pattern and a compensatory lumbar curve. Major thoracolumbar and lumbar curves are less frequent⁴. The three-dimensional nature of the deformity results in torsion in the spine that is most significant at the apex of the major curve, and changes in the structures of the vertebrae and ribs may develop as the curve worsens or progresses⁴. Children with thoracic curves, with larger curves at first presentation, and/or with greater remaining growth potential are at increased risk of progression, but this risk decreases sharply after skeletal maturity⁴. Sex is a recognized risk factor for AIS, with girls having at least a five-fold greater risk of progressive deformity requiring treatment compared to boys⁵. This well-documented sexual dimorphism has prompted speculation that levels of circulating endocrine hormones, particularly estrogen, are important exposures in AIS susceptibility⁶.

The genetic architecture of human AIS is complex, and underlying disease mechanisms remain uncertain. Heritability studies of Northern European^{7,8}, North American^{9,10}, and South Asian¹¹ ancestral groups suggest that disease risk is multifactorial, caused by genetic and environmental contributions^{2,12}. Accordingly, population-based genome-wide association studies (GWAS) in multiple ancestral groups have identified several AIS-associated susceptibility loci, mostly within non-coding genomic regions¹². In particular, multiple GWAS have implicated noncoding regions near the *LBX1*¹³, *ADGRG6* (also known as *GPR126*)¹⁴, and *BNC2*¹⁵ genes. An association with alleles in an enhancer distal to *PAX1*, encoding the transcription factor paired box 1, was primarily driven by females, suggesting that it contributes to the sexual dimorphism observed in AIS¹⁶. Subsequent meta-analysis of combined AIS GWAS identified additional susceptibility loci. These included variants in an intron of *SOX6*, a transcription factor, that along

with *PAX1*, is important in early spinal column formation¹⁷. Furthermore, gene enrichment analyses found significant correlation of AIS-associated loci with biological pathways involving cartilage and connective tissue development¹⁸. A more recent GWAS in a Japanese population identified fourteen additional AIS loci that are candidates for further evaluation¹⁹. In separate studies, genome sequencing in AIS cases and families identified enrichment of rare mutations in the *COL11A2*²⁰ and *HSPG2*²¹ genes, encoding components of the cartilage extracellular matrix (ECM). Hence variation affecting cartilage and connective tissue ECM genes is an emerging theme in the heterogeneous genetic architecture of AIS.

Pre-clinical animal models are essential tools for accelerating mechanistic understanding of AIS and for therapeutic testing¹². In zebrafish, several genetic mutants with larval or later-onset spinal deformity have been described, including *ptk7*^{22,23}, *c21orf59*²⁴, *ccdc40*²⁵, *ccdc151*²⁶, *dys1c1*, and *kif6*²⁷. In mouse, conditional deletion of *Adgrg6* in skeletal cartilage (using *Col2a1*-Cre) produces a progressive scoliosis of the thoracic spine during postnatal development that is marked by herniations within the cartilaginous endplates of involved vertebrae. Progressive scoliosis, albeit to a lesser extent, was also observed when *Adgrg6* was deleted from committed chondrocytes (using ATC-Cre)^{28–30}. These studies demonstrate that cartilage and possibly other osteochondroprogenitor cells contribute to the scoliosis phenotype in these models²⁹. Taken together, genetic and functional studies in mouse, although limited, support the hypothesis that deficiencies in biogenesis and/or homeostasis of cartilage, intervertebral disc (IVD), and dense connective tissues, undermine the maintenance of proper spinal alignment during the adolescent growth spurt¹².

The combined contribution of reported AIS-associated variants is broadly estimated to account for less than 10% of the overall genetic risk of the disease¹⁹. To address this knowledge gap, we sought to define novel loci associated with AIS susceptibility in genes encoding proteins of the ECM (i.e. the “matrisome”^{31,32}). Here we identify new genetic associations with AIS. Further, our functional assessments support a new disease model wherein AIS-associated genetic variation and estrogen signaling perturb a *Pax1-Col11a1-Mmp3* axis in cartilage GPCs.

Results

Nonsynonymous variants in matrisome genes are associated with increased risk of AIS

The “matrisome” has been defined as “all genes encoding structural ECM components and those encoding proteins that may interact with or remodel the ECM”³³. Proteins comprising the global ECM as currently defined have been identified by both experimental and bio-informatic methods³¹. We assembled 1027 matrisome genes as previously identified³⁴, including 274 core-matrisome and 753 matrisome-associated genes (N=1,027 total). For the genes encoding these 1,027 proteins, we identified all nonsynonymous common variants (MAF > 0.01) queried by the Illumina HumanCoreExome-24v1.0 beadchip and determined their genotypes in a discovery cohort of 1,358 cases and 12,507 controls (**Table 1**). After applying multiple quality control measures (see Methods section), we retained 2,008 variants in 597 matrisome genes for association testing (**Supplemental Table 1**). For controls, we utilized 12,507 non-AMD GRU (non-age-related macular degeneration general research use) subjects of European ancestry downloaded from dbGaP (<https://www.ncbi.nlm.nih.gov/gap/>). This sample size was estimated to provide at least 80% power to detect significant associations at the matrisome-wide level ($\alpha \leq 2.5 \times 10^{-5}$), for alleles with population frequency ≥ 0.05 and OR ≥ 1.5 (**Figure 1 – figure supplement 1a**). Two nonsynonymous variants, in *COL11A1* (rs3753841; NM_080629_c.4004C>T; p.(Pro1335Leu); odds ratio (OR)=1.236 [95% CI=1.134–1.347], $P=1.17 \times 10^{-6}$) and *MMP14* (rs1042704; NM_004995_c.817G>A; p.(Asp273Asn); OR=1.239 [95% CI=1.125–1.363], $P=1.89 \times 10^{-5}$) were significantly associated with AIS (**Figure 1a**). Given the sexual dimorphism in AIS and our prior observation of a female-

predominant disease locus¹⁶, we tested the 2,008 variants separately in females (N=1,157 cases and 7,138 controls). In females, the association with rs3753841 remained statistically significant, whereas rs1042704, near *MMP14*, was not associated with AIS in females (**Figure 1 – figure supplement 1b**). Our study was not sufficiently powered to test males separately.

To validate these results, we sought to replicate the associations of rs3753841 and rs1042704 in four independent AIS case-control cohorts, from North America, Europe, and eastern Asia, representing multiple ethnicities (total N = 9,161 AIS cases, 80,731 healthy controls, **Table 1**). Genotypes for both variants were extracted from these datasets and tested for association by meta-analysis together with the discovery cohort (see Methods). Meta-analysis of all cohorts together increased the evidence for association of both variants with AIS risk (**Figure 1b**). While a similar effect size was noted for rs1042704 in Japanese and Han Chinese cohorts, the results were less significant, likely due to lower minor allele frequencies (East Asian MAF = 0.02 compared to total non-Asian cohort MAF = 0.20) in these populations (**Figure 1 – figure supplement 1c**). Plotting recombination across both regions suggested that these signals were likely confined to blocks of linkage disequilibrium within the *COL11A1* and *MMP14* genes, respectively (**Figure 1 – figure supplement 1d, e**).

Although low-resolution structures currently available for collagen triple helices are not useful for modeling the stability change of individual mutations, we noted that the AIS-associated variant P1335L occurs at the third position of a Gly-X-Y repeat and consequently could be structurally important in promoting stability of the triple helix, particularly if it is hydroxylated. Rare dominant mutations in *COL11A1*, often disrupting a Gly-X-Y sequence, can cause Marshall (MRSHS) (OMIM# 154780) or Stickler syndromes (STL2) (OMIM# 604841) marked variously by facial anomalies, sensorineural hearing loss, short stature, spondyloepiphyseal dysplasia, eye anomalies, ectodermal features, and scoliosis. Notably, our AIS cohort and particularly individuals carrying the rs3753841 risk allele were negative for co-morbidities or obvious features of Marshall or Stickler syndromes. Thus, AIS is independently associated with the *COL11A1*^{P1335L} predicted variant.

COL11A1 is expressed in adolescent spinal tissues

We elected to further characterize *COL11A1* in postnatal spine development. *COL11A1* encodes one of three alpha chains of type XI collagen, a member of the fibrillar collagen subgroup and regulator of nucleation and initial fibril assembly, particularly in cartilage³⁵. Spinal deformity is well-described in *Col11a1*-deficient (*cho/cho*) embryos³⁶. In mouse tendon, *Col11a1* mRNA is abundant during development but barely detectable at three months of age³⁷. We analyzed RNA-seq datasets derived from adolescent human spinal tissues³⁸, finding that *COL11A1* was upregulated in cartilage relative to bone and muscle. In cartilage, *PAX1* and *COL11A2*, showed the strongest expression levels relative to other published human AIS-associated genes^{14,16,18,20,21,39} (**Figure 2**). In all, the majority of AIS-associated genes showed the strongest expression levels in cartilage relative to other adolescent spinal tissues.

We next sought to characterize *Col11a1* expression in spines of postnatal mice. To detect Col11a1 protein (collagen α1(XI)), we performed immunohistochemistry (IHC) and immunofluorescence (IF) microscopy using a collagen α1(XI) reactive antibody⁴⁰ in newborn (P0.5) and adolescent (P28) mice. In spines of P0.5 mice, strong staining was observed in the nucleus pulposus (NP) and in surrounding annulus fibrosus (AF) (**Figure 2b**). In P28 mice, the compartments of the IVD were more distinct, and strong collagen α1(XI) staining was observed in each (**Figure 2c**). In regions of the cartilage endplate (CEP)-vertebral bone transition, collagen α1(XI) was detected in columnar chondrocytes, particularly in the hypertrophic zone adjacent to condensing bone (**Figure 2c**). We also examined collagen α1(XI) expression in ribs, as these structures are also involved in the scoliotic deformity⁴¹. In P28 rib growth plates, as in spine, a biphasic pattern was observed in which collagen α1(XI) reactivity was most pronounced around cells of the presumed

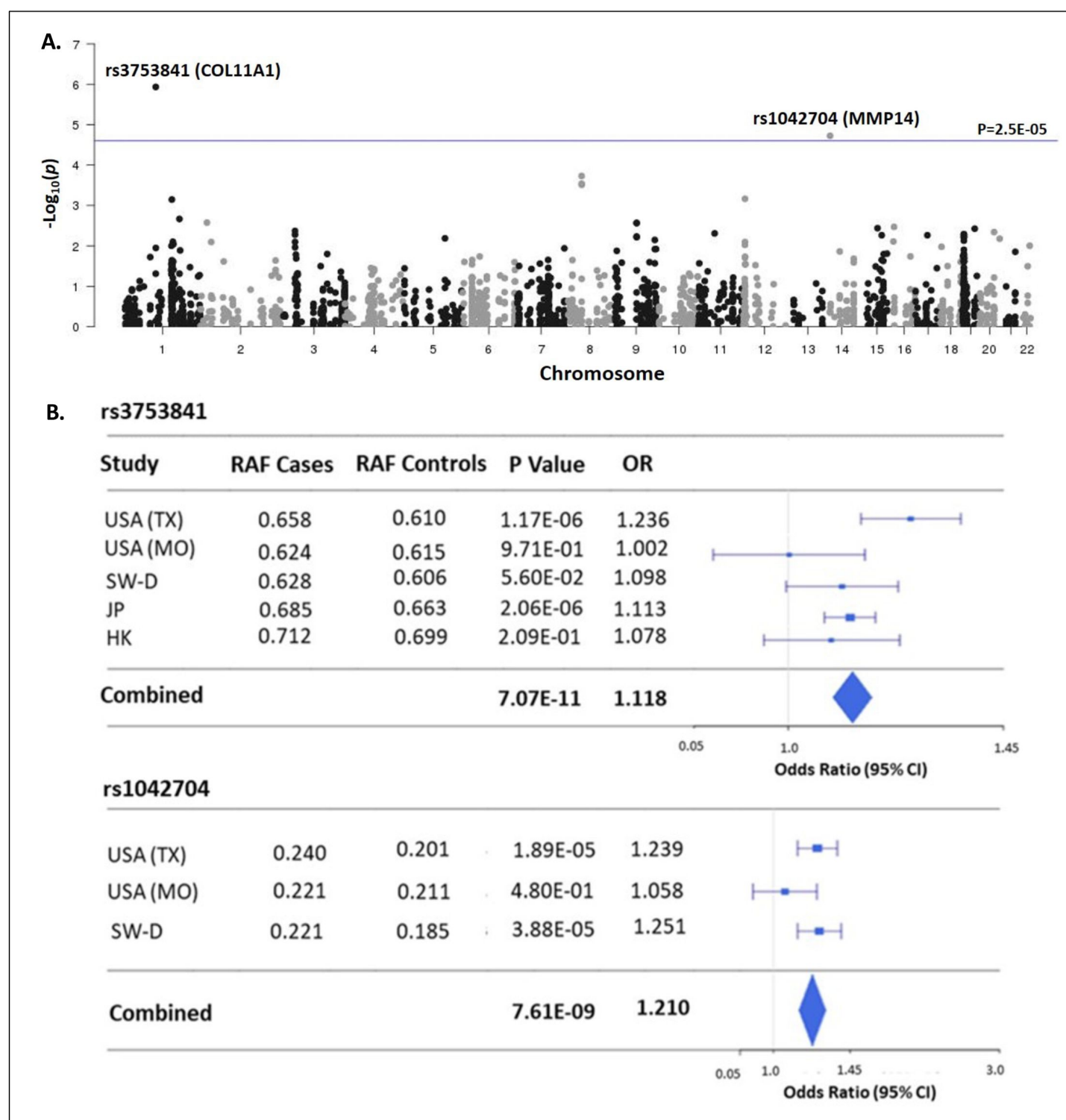


Figure 1.

Matrisome-wide association study.

a. Manhattan plot showing $-\log_{10} P$ -values (y axis) versus chromosomal position (x axis) for the 2,008 common coding variants tested in the discovery study USA (TX). The horizontal line represents the threshold for significance level (P -value $< 2.5 \times 10^{-5}$) after Bonferroni multiple testing correction. **b.** Tests of association for SNPs rs3753841 and rs1042704 in discovery and independent replication cohorts. RAF reference allele frequency; OR odds ratio; CI confidence interval.

Cohort	Ethnicity	Stage	Subjects	Cases		Controls	
				Male	Female	Male	Female
USA (TX)	NHW	Discovery	13,865	201	1,157	5,369	7,138
USA (MO)	NHW	Replication	2,951	201	1,012	1,049	689
SW-D	NHW	Replication	4,627	222	1,409	505	2,491
JP	EAS (Japanese)	Replication	79,211	323	5,004	40,205	33,679
HK	EAS (Han Chinese)	Replication	3,103	178	812	858	1,255
Total			103,757		10,519		93,238

Table 1.

Participating studies.

NHW: Non-Hispanic White. EAS: East Asian.

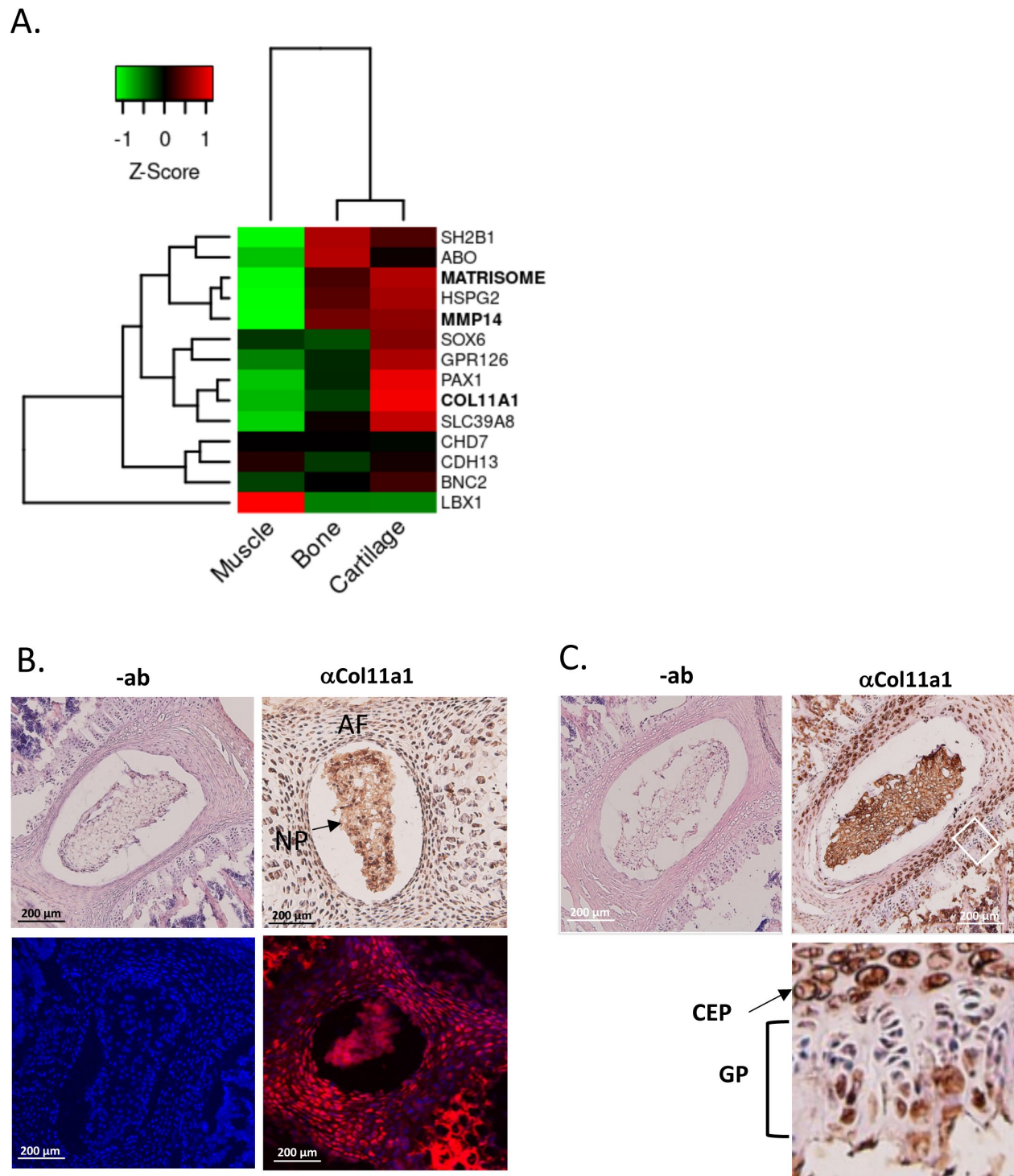


Figure 2.

***Col11a1* and *Mmp14* expression in spine.**

- a)** A heatmap of transcript per million (TPM) values of *COL11A1*, *MMP14*, and other published genes associated with AIS. The average TPM value of matrisome genes is represented as MATRISOME.
- b)** Localization of collagen type XI alpha chain 1 in P0.5 mouse IVD. Immunohistochemistry (IHC) and immunofluorescence (IF) staining in IVD at day P0.5. Negative antibody controls in IVD are shown at left; antibody-positive IHC and IF are shown at right.
- c)** Localization of collagen type XI alpha chain 1 in P28 mouse IVD. Negative antibody IHC control shown at left; antibody positive IHC shown at right. Enlarged, rotated view of white boxed area shows a biphasic staining pattern. CEP – cartilage endplate; GP – growth plate.

resting and pre-hypertrophic/hypertrophic zones (**Figure 2 – figure supplement a,b**). These data show that in mouse, collagen $\alpha 1(\text{XI})$ is detectable in all compartments of young postnatal IVD and is particularly abundant in the chondro-osseous junction region of IVD and vertebral growth plate.

Col11a1 is positively regulated by Pax1 in mouse spine and tail

We previously identified AIS-associated variants within a putative enhancer of *PAX1* encoding the transcription factor Paired Box 1^{16,18}. *Pax1* is a well-described marker of condensing sclerotomal cells as they form segments that will eventually become the IVD and vertebrae of the spine^{41–43}. We generated *Pax1* knock-out mice (*Pax1*^{−/−}) using CRISPR-Cas9 mutagenesis and validated them using sequencing and Southern blot (**Figure 3-figure supplement 3a-d**). Homozygous *Pax1*^{−/−} mice were viable and developed kinks in the tail, similar to other *Pax1* deficient mice⁴⁴. We next compared the expression of collagen $\alpha 1(\text{XI})$ protein in IVD and condensing bone of wildtype and *Pax1*^{−/−} mice by performing IF staining in P28 spines (**Figure 3a**). In wildtype IVD, strong overlapping expression of collagen $\alpha 1(\text{XI})$ and *Pax1* cells was observed, mostly within the CEP and chondro-osseous interface (**Figure 3a**). The IVD in *Pax1*^{−/−} mice was highly disorganized, without discernable NP, AF, and CEP structures as has been reported⁴⁵. Cells within the IVD were negative for *Pax1* staining as expected, and collagen $\alpha 1(\text{XI})$ staining was also clearly reduced in CEP and the chondro-osseous vertebral border. To test the effect of *Pax1* on expression of *Col11a1* and other AIS-associated genes during embryonic development, RNA was isolated from vertebral tissue dissected from the tails of E12.5 wild-type and *Pax1*^{−/−} mice and subjected to bulk RNA-seq and quantitative real-time PCR (qRT-PCR). Gene-set enrichment analysis of RNA-seq was most significant for the gene ontology term “extracellular matrix” (**Figure 3b**). Interestingly, the AIS-associated gene *Adgrg6* was amongst the most significantly dysregulated genes in the RNA-seq analysis (**Figure 3c**). By qRT-PCR analysis, expression of *Col11a1*, *Adgrg6*, and *Sox6* were significantly reduced in female and male *Pax1*^{−/−} mice compared to wild-type mice (**Figure 3d-g**). These data support a mechanistic link between *Pax1* and *Col11a1*, and the AIS-associated genes *Gpr126* and *Sox6*, in affected tissue of the developing tail.

Col11a1 regulates Mmp3 expression growth plate chondrocytes

COL11A1 has been linked with ECM remodeling and invasiveness in some cancers⁴⁶. In solid tumors, *COL11A1* has been shown to alter ECM remodeling by enhancing *MMP3* expression in response to *TGF β* ⁴⁶. *MMP3* encodes matrix metalloproteinase 3, also known as stromelysin, an enzyme implicated in matrix degradation and remodeling in connective tissues⁴⁷. We confirmed strong *MMP3* mRNA expression, relative to *COL11A1*, in human spinal cartilage and bone, but minimal expression in spinal muscle (**Figure 4 – figure supplement 4a**). We next generated *Col11a1*-deficient chondrocytic cell cultures from rib growth plate chondrocytes (GPCs) of P0.5 *Col11a1*^{fl/fl} mice⁴⁰ by treating with Cre-expressing adenoviruses. After confirming *Col11a1* excision (**Figure 4a**), we compared *Mmp3* expression in these cells to cells treated with GFP expressing adenoviruses lacking Cre activity. We found that *Mmp3* expression was significantly increased in GPCs where *Col11a1* mRNA expression was downregulated by about 70% (**Figure 4b**). Furthermore, Western blotting in these cells demonstrated a ~2-5-fold increase in pro-, secreted, and active forms of *Mmp3* protein when collagen $\alpha 1(\text{XI})$ was reduced. The proteolytic processing *per se* of precursor *Mmp3* into active forms⁴⁸ did not appear to be affected by *Col11a1* expression (**Figure 4c**). These results suggest that *Mmp3* expression is negatively regulated in *Col11a1* in mouse GPCs.

To test whether *Col11a1* affects *Mmp3* levels *in vivo*, we bred *Col11a1*^{fl/fl} female mice with *Col11a1*^{fl/fl} ATC males carrying the Acan enhancer-driven, doxycycline-inducible Cre (ATC) transgene⁴⁹. ATC has been shown to harbor Cre-mediated recombination activity in most differentiated chondrocytes and in nucleus pulposus within two days of treating pregnant mothers with doxycycline starting at E15.5⁴⁹. ATC activity was confirmed by crossing this line to

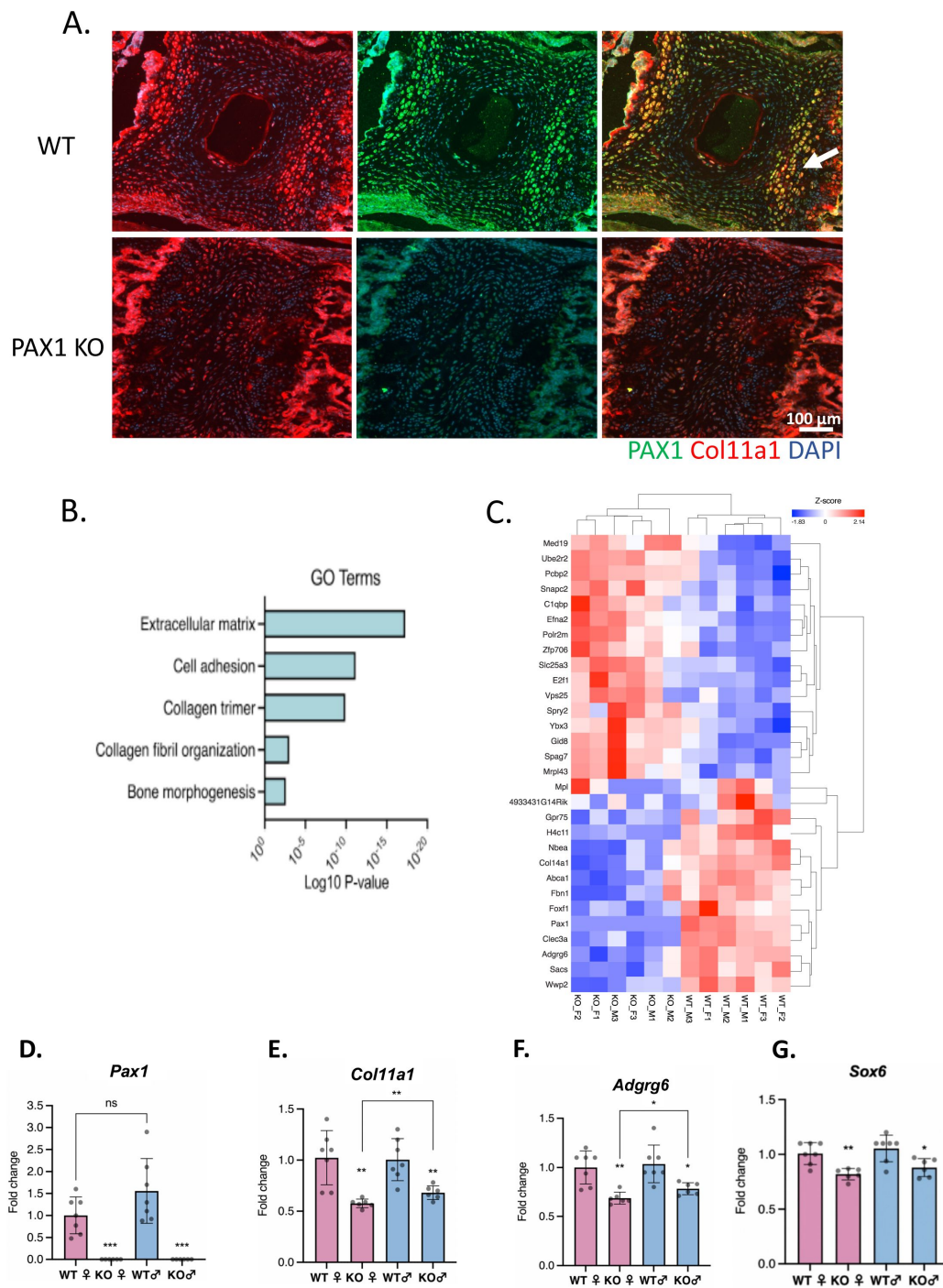


Figure 3.

Assessing *Pax1* regulation of *Col11a1*.

- a)** IF staining of P28 IVD using Pax1 (green) and Col11a1 specific (red) antibodies with DAPI nuclear counterstain. Arrow denotes cells at the CEP-condensing bone interface of IVD that show positive staining for both Pax1 and Col11a1.
- b)** Gene ontology (GO) analysis of differentially expressed genes in E12.5 tail WT and Pax1-null mice.
- c)** Heatmap of differentially expressed genes (P-value<0.0001) in E12.5 tails of WT and Pax1-null mice
- d-g)** Gene expression levels dissected from E12.5 mouse tail from wild type (WT) and Pax1-null (KO) mice as determined by qRT-PCR. Each value represents the ratio of each gene expression to that of β -actin, and values are mean $\pm$ standard deviation. The expression value of WT female group was arbitrarily set at 1.0. Each dot represents one embryo and statistical differences were determined using a two-sided unpaired t-test (P*<0.05, P**<0.01, P***<0.001).

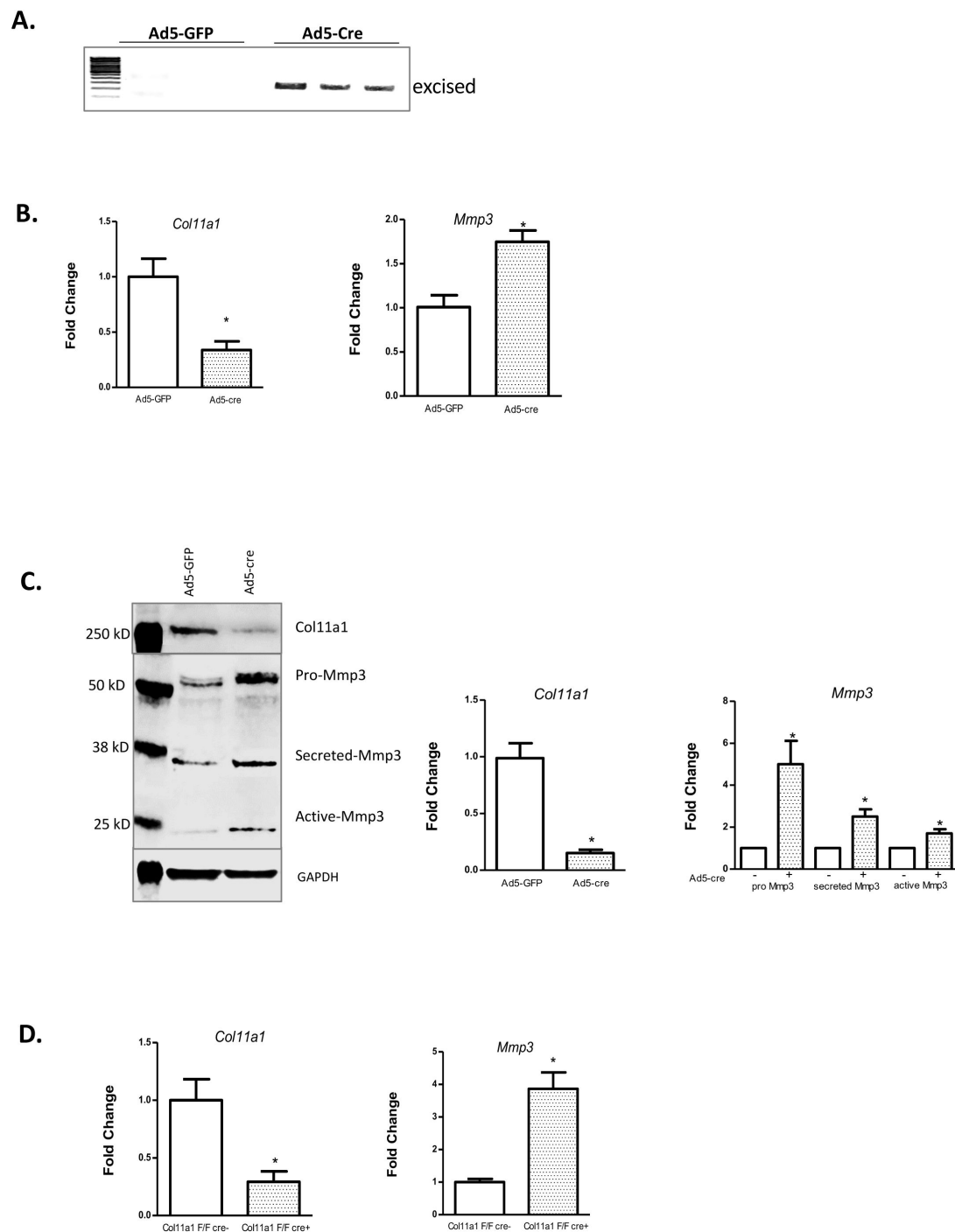


Figure 4.

Col11a1 regulation of *Mmp3* expression in cartilage.

- a)** PCR assay of *Col11a1* excision in *Col11a1^{fl/fl}* cultured rib cartilage cells.
- b)** Gene expression levels from *Col11a1^{fl/fl}* cultured rib cartilage cells transduced with GFP (Ad5-GFP, left) or Cre expressing adenovirus (Ad5-cre, right) as determined by qRT-PCR. Each value represents the ratio of each gene expression to that of *GAPDH*, and values are mean $\pm$ standard deviation. The expression value of control Ad5-GFP results ($N \geq 4$ experiments) was arbitrarily set at 1.0. Statistical differences were determined using a two-sided unpaired t test ($P^* < 0.05$).
- c)** Representative Western blot of cultured cartilage cells after Ad5-GFP or Ad5-cre transduction. Protein size ladder is shown in lane 1. Quantification of bands detected by Western blotting, where scramble results were set to 1.0 from $N=4$ independent experiments is shown at right.
- d)** Gene expression levels from dissected *Col11a1^{fl/fl}* ATC rib cartilage, analyzed as described in a). Results shown are from $N=3$ mice.

the R26td^[Tomato] reporter that ubiquitously expresses the fluorescent gene Tomato after Cre recombination. Strong Cre activity was seen in P0 pups of mothers treated with doxycycline at E15.5 in the NP, CEP, and annulus fibrosus (AF) of the IVD and in chondrocytes of the growth plates (**Figure 4 – figure supplement 4b**). Pregnant *Col11a1*^{fl/fl} females were treated with doxycycline water from E15.5 to induce Cre expression in differentiated chondrocytes. Excision of *Col11a1* was confirmed in DNA from costal cartilage of *Col11a1*^{fl/fl}ATC cre-positive offspring (**Figure 4 – figure supplement 4c**). Consistent with results obtained by *in vitro* excision of *Col11a1*, cartilage from mice deficient in *Col11a1* showed ~4-fold upregulation of *Mmp3* mRNA expression relative to *Col11a1*^{fl/fl} mice (**Figure 4d** [↗](#)).

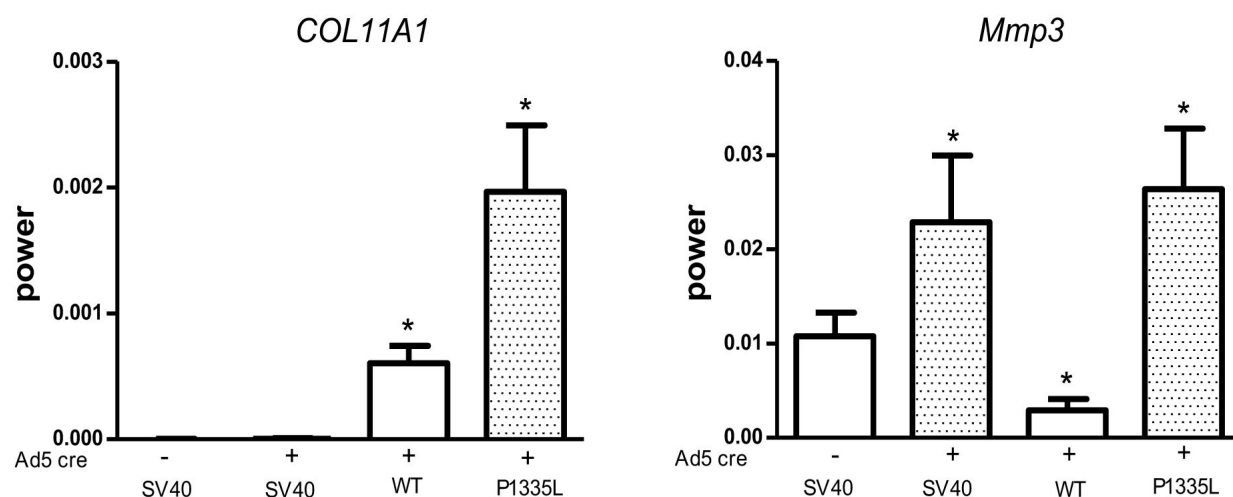
AIS-associated variant in COL11A1 perturbs its regulation of MMP3

COL11A1 missense mutations have been shown to evoke transcriptional changes in ECM genes in cancer cells⁵⁰ [↗](#). We therefore tested whether the *COL11A1*^{P1335L} sequence variant alters its regulation of *Mmp3* in cartilage chondrocytes. For this, SV40-immortalized cell lines were established from *Col11a1*^{fl/fl} mouse rib cartilage and transduced with lentiviral vectors expressing green fluorescent protein (GFP) and *COL11A1*^{wt}, *COL11A1*^{P1335L}, or vector alone. After transduction, GFP-positive cells were grown to 50% confluence and treated with Cre-expressing adenovirus to remove endogenous mouse *Col11a1* (**Figure 5a,c** [↗](#)). Using a human-specific *COL11A1* qRT-PCR assay, we detected overexpression of *COL11A1*^{wt} and *COL11A1*^{P1335L} compared to untransduced cells regardless of Cre expression (**Figure 5a** [↗](#)). Western blotting with an antibody directed against the HA epitope tag confirmed also confirmed overexpression of human *COL11A1* (**Figure 5c** [↗](#)). Endogenous *Mmp3* upregulation was evident by both qRT-PCR and Western blotting in untransduced cells treated with Ad5 Cre, as expected, and the reverse was observed in cells overexpressing human wildtype *COL11A1*, consistent with the negative regulation we previously observed (**Figure 5 b,c** [↗](#)). However, the *COL11A1*^{P1335L} mutant failed to downregulate *Mmp3* expression despite being overexpressed (**Figure 5b,c** [↗](#)). Thus, regulation of *Mmp3* is perturbed in the presence of the *COL11A1*^{P1335L} variant in these cells.

Esr2-Col11a1-Mmp3 form a signaling axis in cartilage cells

The expression of *Col11a1*, and of other ECM genes, is known to be estrogen-responsive in certain tissues, such as ovarian follicular cells⁵¹ [↗](#). Because of the suspected role of endocrine hormones in AIS, we investigated whether *Col11a1* expression was responsive to estrogen treatment and/or levels of estrogen receptor following siRNA-mediated knockdown experiments in cultured cartilage GPCs. We first validated that *Mmp3* levels were significantly increased after *Col11a1* knockdown (**Figure 6a** [↗](#)). Estrogen receptor 2 (*Esr2*), but not estrogen receptor alpha (*Esr1*), was detected in mouse GPCs by qRT-PCR (data not shown). We next tested the consequences of *Col11a1* and *Esr2* siRNA-mediated knockdown on gene expression in cartilage GPCs. After *Esr2* knockdown, *Col11a1* as well as *Pax1* were significantly upregulated compared to scramble control, while *Mmp3* expression was significantly downregulated (**Figure 6b** [↗](#)). *Pax1* expression was also upregulated by *Col11a1* knockdown alone. These results suggest a negative feedback loop between *Pax1* and *Col11a1* in these cells. Simultaneous knockdown of *Col11a1* and *Esr2* expression reduced *Mmp3* expression to normal levels, supporting a possible interaction between *Col11a1* and estrogen signaling in regulating *Mmp3*. Treating GPCs with tamoxifen, an estrogen receptor modulator, also upregulated *Col11a1* expression to similar levels as observed after *Esr2* knockdown, compared to cells treated with DMSO carrier (**Figure 6 figure supplement 6a**). These results further support estrogen signaling in regulating *Col11a1* expression. In cultured rat CEP cells, *Esr2* mRNA was downregulated, and *Mmp3* mRNA was upregulated after *Col11a1* knockdown, as observed in mouse GPCs (**Figure 6c** [↗](#), **Figure 6 – figure supplement 6b**). However, *Esr2* knockdown did not significantly impact *Col11a1* or *Mmp3* expression in these cells (**Figure 6c** [↗](#)). Hence, we conclude that in mouse GPCs, expression of the ECM-clearing enzyme

A.



B.

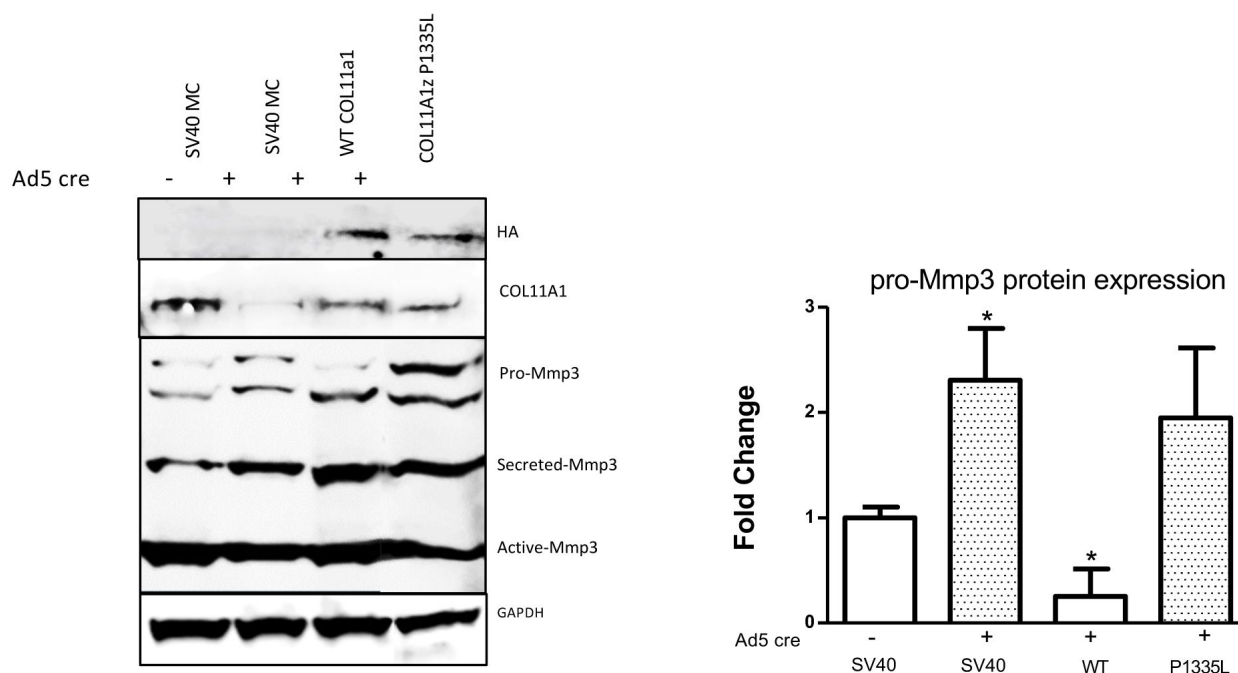


Figure 5.

***Col11a1*^{P1335L} regulation of *Mmp3* expression in lentiviral transduced mouse GPCs.**

a) qRT-PCR of human *COL11A1* and endogenous mouse *Mmp3*. "SV40" refers to immortalized mouse chondrocytes transduced with the lentiviral vector only. Significant fold changes ($P \leq 0.5$) relative to vector-only transfected cells are shown by *.

b) Western blot corresponding to experiments shown in (a) using antibodies to detect Col11a1, HA epitope tag representing human Col11a1 only, endogenous Mmp3, and GAPDH. Quantification at right shows Mmp3 expression in each experiment relative to GAPDH.

Mmp3 gene is normally repressed by collagen $\alpha 1(XI)$, but this regulation is disrupted by AIS-associated variation in *COL11A1*. Further, we find that estrogen signaling also dysregulates the *Col11a1-Mmp3* interaction.

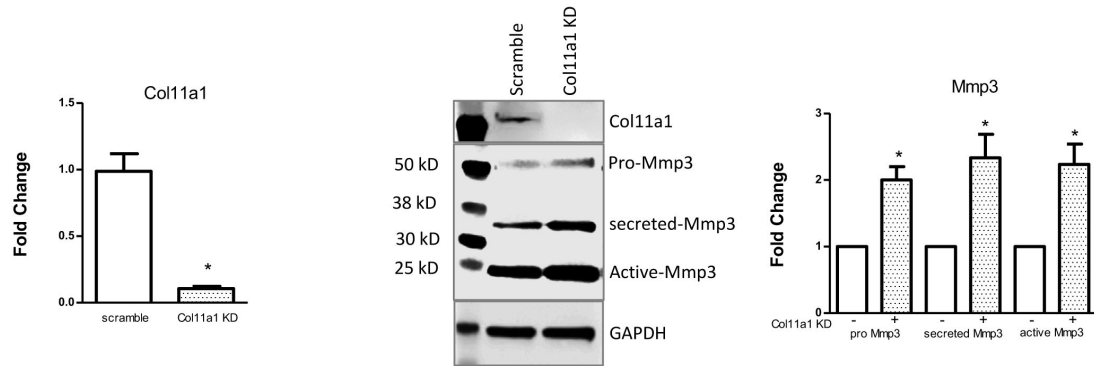
Discussion

Adolescent idiopathic scoliosis has been described in the medical literature for centuries, yet its underlying etiology has remained enigmatic⁵². Given that AIS originates in children who appear to be otherwise healthy, even its tissue of origin has been difficult to discern, and long debated¹². The advent of powerful genotyping and sequencing methods in the last two decades has led to breakthrough discoveries of genetic loci associated with AIS, most in non-coding regions of the genome that are difficult to interpret biologically¹². Aggregating these results, however, provided supportive evidence that pathways of cartilage and connective tissue ECM development are relevant in AIS etiology^{12, 18}. Here, in the largest multi-ethnic human cohort studied to date, we elected to test the hypothesis that alterations in ECM proteins themselves contribute to AIS susceptibility. This approach yielded most significant evidence for a common protein-altering variant in the *COL11A1* gene encoding collagen $\alpha 1(XI)$, a minor yet critical component of cartilaginous ECM. Moreover, our studies define a *COL11A1*-mediated disease pathway (**Figure 7**) and point to the chondro-osseous junction of IVD and vertebrae spine as a relevant cellular compartment in AIS etiology.

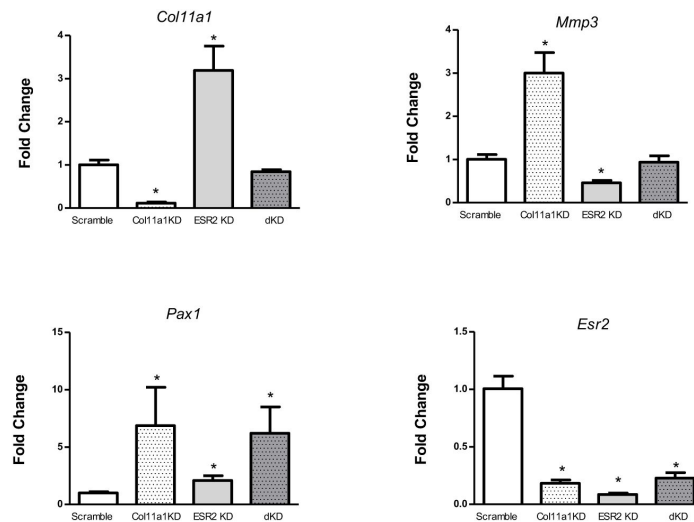
The results of this study together with the previous observation of *COL11A2* rare variant enrichment in AIS supports a role for the collagen $\alpha 1(XI)$ heterotrimer itself in its pathogenesis²⁰. Collagen type XI, composed of three chains encoded by the *COL11A1*, *COL11A2*, and *COL2A1* genes (OMIM #s 120280, 120290, 120140, respectively), is a minor component of collagen type II fibrils that are abundant in cartilage. Collagen type XI is also broadly expressed in testis, trachea, tendons, trabecular bone, skeletal muscle, placenta, lung, brain neuroepithelium, the vitreous of the eye, and intervertebral discs⁵³. In the pericellular space, collagen $\alpha 1(XI)$ initiates fibrillogenesis with collagen type II fibrils, maintaining regular spacing and diameter of the collagen fibrils, while organizing the pericellular surface by interaction with cartilage proteoglycans^{54, 55}. Purified human collagen type XI, when added back to chondrocytes in *in vitro* culture, stimulates chondrogenesis while inhibiting hypertrophy, as measured by histologic staining, proliferation assays, and relative expression of chondrogenic early marker genes⁵⁶. In newborn and one-month old mice, we found that collagen $\alpha 1(XI)$ was abundant in IVD and at the chondro-osseous junction of IVD and vertebrae, particularly concentrated in pre hypertrophic/hypertrophic chondrocytic cells. In long bone growth plates, Long et al.⁵⁷ recently identified eight distinct cell clusters after unsupervised analysis of single cell (scRNAseq) of flow sorted hypertrophic chondrocytes from Col10a1Cre;Rosa26fs-tdTomato mice. At embryonic stage 16.5 (E16.5), *Col11a1* expression was highest in cells with signatures of pre-hypertrophic to hypertrophic transition, and lowest in cells with osteogenic signatures (M. Hilton, personal communication)⁵⁷. Taken together, these results suggest that collagen $\alpha 1(XI)$ normally participates in maintaining growth plate cells in a hypertrophic, pre-osteogenic state, although little is known about its precise molecular function in that compartment, or in the IVD, during spinal development. Spines of *Col11a1*-deficient mice (*cho/cho*) show incompletely formed vertebral bodies, spinal curvatures, and decreased separation between vertebrae, which are themselves less mineralized than in wildtype mice³⁶. Notably, common *COL11A1* variants also have been associated with adult lumbar disc herniation (LDH) and degeneration (LDD), as well as DXA measured bone size, spinal stenosis and spondylolisthesis^{58–60}.

Pax1 is a well-described marker of early spine development, where it activates a gene expression cascade starting at E12.5–13.5 in mouse development^{44, 61, 62}. Our data provided evidence that *Pax1* regulates expression of *Col11a1*, as well as the AIS-associated genes *Sox6* and *Gpr126*, in E12.5 tails of both male and female mice. The reduction of *Col11a1* is consistent with a prior study

A.



B.



C.

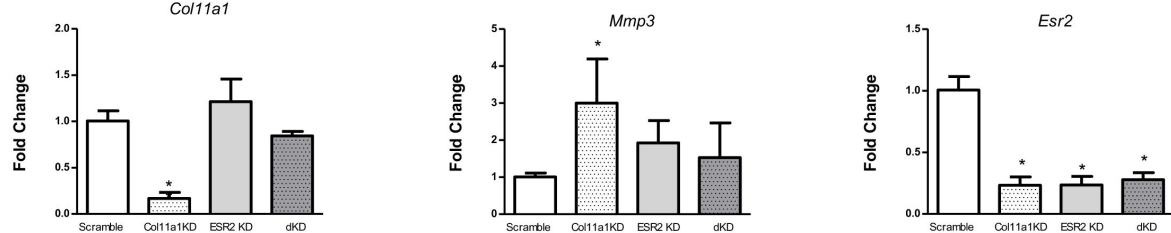


Figure 6.

Effects of estrogen receptor beta on *Col11a1* *Mmp3* signaling axis.

- a)** Representative Western blot of cultured cartilage cells after scramble or *Col11a1*-specific siRNA knockdown. Protein size ladder is shown in lane 1. Quantification of bands detected by Western blotting, where scramble results were set to 1.0 from N=4 independent experiments is shown at right.
- b)** Gene expression levels of *Col11a1* (left), *Mmp3* (middle), and *Esr2* (right) mRNA in cultured rib cartilage cells showing fold change relative to the scramble control. ddk = double *Col11a1*-*Esr2*-specific siRNA knockdowns. Results are representative of n≥3 experiments.
- c)** Gene expression levels from rat CEP cells, as described in (b).

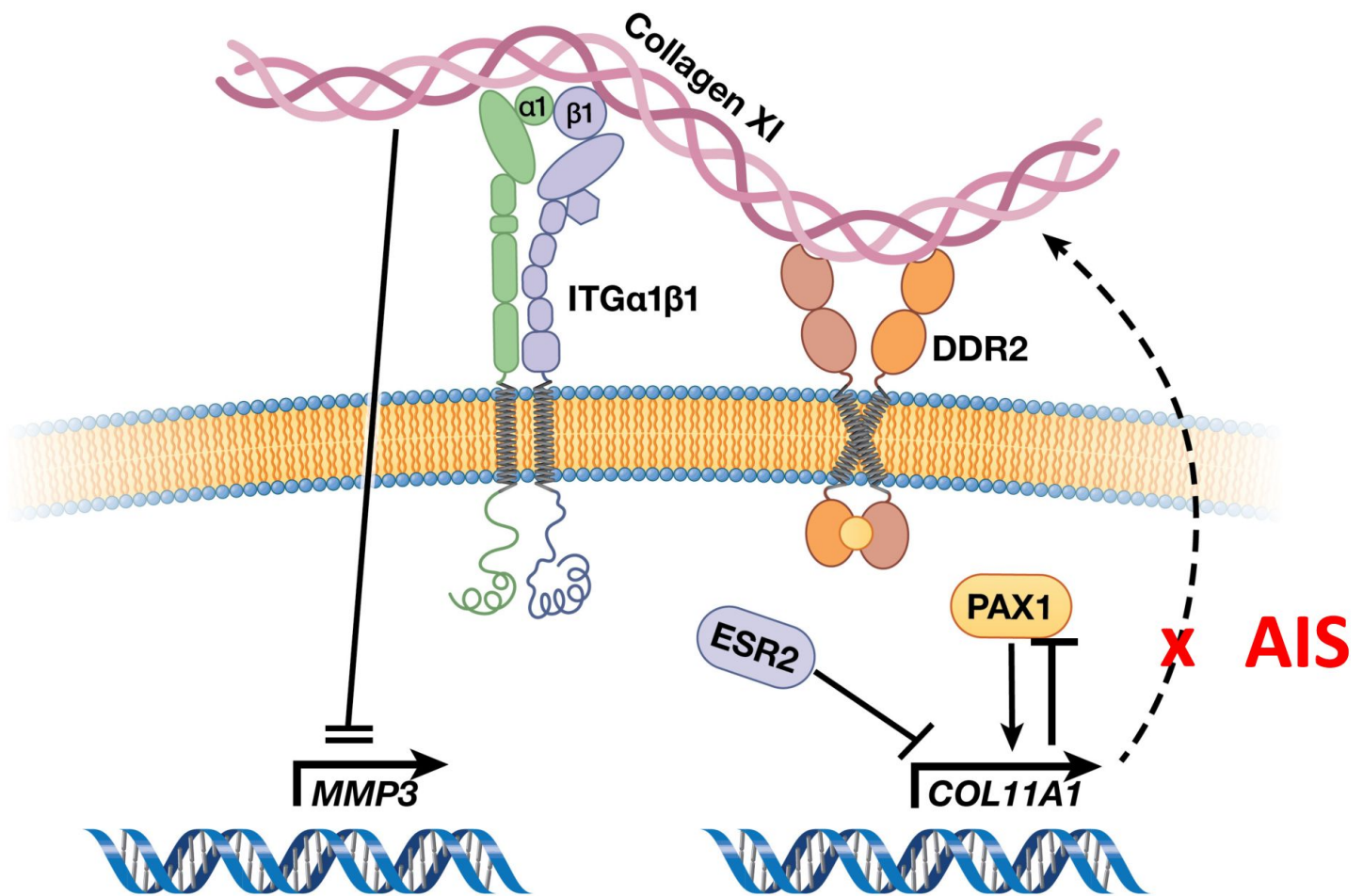


Figure 7.

Cartoon depiction of a collagen XI-mediated signaling axis in cartilage. Collagen XI is held in the pericellular space by integrins and DDR2. COL11A1, under the regulation of ESR2 and PAX1, signals through unknown mechanisms and inhibits MMP3 transcription.

of gene expression in flow-sorted GFP-labeled Pax1^{-/-} embryonic IVD cells⁶². It is likely that *Col11a1* expression in developing tail is directly activated by binding Sox transcription factors, as a prior genomic study using chromatin immunoprecipitation and sequencing (ChIP-seq) in rat chondrosarcoma cells identified super enhancers near the *Col11a1* gene that were bound multiple times by the Sox9 and Sox6⁶³. The Sox5/6/9 trio is known to regulate many of the same genes as Pax1⁶², but whether it mediates the Pax1 regulation of *Col11a1* observed in our study is unknown.

In mouse postnatal spines, we observed co-localization of collagen α1(XI) and Pax1 protein specifically within the cartilaginous endplate-vertebral junction region that includes the vertebral growth plate. The endplate, which is important as the site allowing diffusion of nutrients from the circulation into the avascular inner IVD, harbors subpopulations of cells expressing type II collagen presumably organized by collagen type XI^{41,43}. While the endplate is continuous with the vertebral growth plate in mice, it is important to note that in humans the endplate and vertebrae become distinctly separate structures with closure of the growth plates at puberty⁴¹. This is also the site of the ring apophyses that form the insertion of the IVD into vertebrae⁶⁴. Lagging maturity of the ring apophysis, combined with mechanical forces across the IVD in the horizontal plane, has been proposed as an initiating factor leading to rotatory decompensation in the adolescent spine in AIS^{64,65}. Recently, Sun et al. reported the discovery of a vertebral skeletal stem cell (vSSC) residing in the endplate and marked by expression of the genes *Zic1* and *Pax1*, along with other cell surface markers⁶⁶. These vSSCs also express high levels of *Col11a1* (M. Greenblatt, personal communication). It is interesting to consider that AIS-associated genetic variation, perhaps together with mechanical forces, could alter the differentiation trajectory of this cell niche. Altogether, extant data and our results strongly suggest that cell populations at the IVD-vertebral junction region are relevant in AIS pathogenesis. Further investigation is warranted to understand the developmental programmes of cells in this region of the spine.

Matrix metalloproteinase 3, also known as stromelysin, is a secreted enzyme expressed in connective tissues and in regions of endochondral ossification⁶⁷. MMP3 has degradative activity toward a variety of ECM components, including proteoglycans, fibronectin, laminin, but notably not type I collagen⁶⁸. Additionally, in chondrocytes MMP3 also has been shown to translocate to the nucleus, where it activates transcription of connective tissue growth factor (*CTGF/CCN2*) by binding to an element known as transcription enhancer dominant in chondrocytes (TRENDIC)^{69,70}. Our observations of a *Col11a1-Mmp3* signaling axis in GPCs and CEP cells raise the possibility that *Col11a1* variation may have consequences for both MMP3 enzymatic activity and for *MMP3* mediated transcriptional programming in these cells. *COL11A1* missense mutations, usually altering glycine or proline in Gly-X-Y repeats in the collagen α1(XI) helical domain as with *COL11A1*^{P1335L}, are reported to be frequent in cutaneous squamous cell carcinomas and have been linked to transcriptional changes and tumor invasiveness⁵⁰. The mechanisms by which GPCs or other cells sense such single amino acid changes in *COL11A1* and induce subsequent transcriptional responses are unknown but may involve direct interaction with integrins in the pericellular space⁵⁰.

We found that *Col11a1* expression is sensitive to estrogen receptor status in GPCs, suggesting that in these cells it is responsive to estrogen signaling. Collagen α1(XI) is also a key player in organizing the pericellular space, which is critical for transmitting mechanical forces from the ECM to the cell⁷¹. Our findings support a new model wherein environmental cues, i.e. mechanical forces and estrogen signaling, intersect with genetic variation in a *Pax1-Col11a1-Mmp3* signaling axis in the IVD-vertebral junction and thereby increase AIS susceptibility.

Discovery study

The cases in the discovery stage (USA TX: n=1,358) were recruited at Scottish Rite for Children as approved by the Institutional Review Board of the University Texas Southwestern Medical Center as previously described⁷². Subjects were genotyped on the Illumina HumanCoreExome BeadChip (Illumina, San Diego, CA, USA). For controls, we utilized 12,507 non-AMD GRU (non-age related macular degeneration general research use) subjects of the European ancestry downloaded from dbGaP web site (<https://www.ncbi.nlm.nih.gov/gap/>) from the International Age-Related Macular Degeneration Genomics Consortium study (IAMDGC: phs001039.v1.p1.c1; https://www.ncbi.nlm.nih.gov/projects/gap/cgi-bin/study.cgi?study_id=phs001039.v1.p1). The subjects from the IAMDGC study were also genotyped on the Illumina HumanCoreExome Beadchip-24v1.0 platform⁷³. We merged cases and controls and applied quality controls to the genotypes for 468,801 overlapping SNPs using PLINK.1.9⁷⁴ as described in¹⁸. In summary, samples with sex inconsistencies or from duplicated or related individuals or ancestral outliers as identified by principal component analysis (PCA) were removed, leaving 13,865 samples in the analysis. Genotypes were corrected for strand direction, and SNPs with call-rate per marker <95%, deviating from Hardy-Weinberg equilibrium (HWE) (cutoff P-value = 10^{-4}), or with significant missingness rate between cases and controls (cutoff P-value = 10^{-4}) were removed, leaving 341,759 SNPs in the analysis. Genotypes for SNPs across autosomal chromosomes were imputed using Minimac3 with the 1000G-Phase3.V.5 reference panel as described in the instructions available from the software website⁷⁵. Protein coding changes were annotated with ANNOVAR using RefSeq-based transcripts⁷⁶. External databases included allele frequencies from gnomAD⁷⁷; variant pathogenicity in Clinvar⁷⁸; Combined Annotation Dependent Depletion (CADD) scores⁷⁹; Genomic Evolutionary Rate Profiling (GERP) scores⁸⁰ and protein domains in IntroPro⁸¹. Only bi-allelic common (MAF>0.01) protein-altering SNPs with imputation quality Rsq >=0.3 within matrisome genes³⁴ were included for further analysis. Matrisome genes used can be found in the Molecular Signature Database (MsigDB)^{82,83} (https://www.gsea-msigdb.org/gsea/msigdb/cards/NABA_MATRISOME). Genetic association for the imputed allele dosages in the discovery cohort (USA TX) was performed in Mach2dat⁸⁴ using logistic regression with gender and ten principal components as covariates. The genomic regions of the associated loci were visualized with LocusZoom software⁸⁵ utilizing linkage disequilibrium (LD) information from 1000 Genomes EUR populations.

Meta-analysis study

For the meta-analysis stage we utilized four cohorts [USA MO: n=2,951(1,213 cases and 1,738 controls), Swedish-Danish populations [SW-D: n= 4,627 (1,631 cases and 2,996 controls), as described in^{18,86}], Japan [JP: n=79,211(5,327 cases and 73,884 controls)] and Hong Kong [HK: n=3,103 (990 cases and 2,113 controls)] to check significant candidates from the discovery study. Summary statistics across the discovery study and the 4 replication cohorts [total N=103,757 (10,519 cases and 93,238 controls)], were combined as previously described¹⁸ using METAL⁸⁷. USA MO cohort: Whole exome sequencing (WES) data from 1,213 unrelated idiopathic scoliosis cases of European ancestry with spinal curvature greater than 10° Cobb angle were derived from the Adolescent Idiopathic Scoliosis 1000 Exomes Study (dbGAP accession number: phs001677), and recruited from St. Louis Children's Hospital, and St. Louis Shriners Hospital for Children. Patients and/or parents provided study consent and IRB approval was obtained from each contributing institution. For controls, exome data from 1,738 unrelated samples of European ancestry were provided by investigators at Washington University School of Medicine in St. Louis, MO (dbGAP accession numbers: phs000572.v8.p4 and phs000101.v5.p1), and Oregon Health & Science University in Portland, OR (<https://gemini.conradlab.org/>). Exome data were aligned to the human genome reference (GRCh37) using BWA-MEM (v0.7.15). Variant calling of single nucleotide variants (SNVs) and insertion and deletion variants (INDELs) were generated first for each single sample in cases and controls and then combining all samples with joint genotyping method,

described in GATK Best-Practices (Genome Analysis Toolkit (GATK v3.5) <https://gatk.broadinstitute.org/hc/en-us/sections/360007226651-Best-Practices-Workflows>). All cases and controls samples were classified as unrelated and of European ancestry using relationship inference⁸⁸ and principal component analysis⁷⁴. Association analysis of variants rs3753841 and rs1042704 were performed using logistic regression adjusting for sex and PCs in PLINK⁷⁴.

JP cohort

Informed consents were obtained from all the subjects or their parents and the ethics committees of RIKEN and participating institutions approved this study. 5,327 case subjects were recruited from collaborating hospitals (Japanese Scoliosis Clinical Research Group) as previously described¹⁹. For controls, 73,884 subjects were randomly selected from the BioBank Japan Project, and subjects were genotyped on Illumina Human BeadChips as previously described^{15,19}. Imputation and association analyses in JP were performed as previously described⁸⁹.

HK cohort

3,103 subjects were recruited at The Duchess of Kent Children's Hospital as approved by the Institutional Review Board of the University of Hong Kong/Hospital Authority Hong Kong West Cluster (IRB approval number: UW 08-158). All 990 cases were characterized by Cobb angles greater than 40 degrees with onset age between 10 and 18 years old. Congenital, neuromuscular and syndromic scoliosis samples were excluded. We used 2,113 controls from the Chinese population with no spinal deformities on MRI scans⁹⁰. Cases and controls were genotyped using the Illumina Infinium OmniZhongHua-8 BeadChip and analyzed with GenomeStudio 2.0 software. The quality control approach adopted the GWA tutorial developed by Andries et al.⁹¹. The filtered genotyping data of cases and controls was phased and imputed using SHAPEIT⁹² and IMPUTE2⁹³, respectively. Logistic model association analysis was performed using PLINK 1.9⁷⁴.

Stratification-by-sex test

To investigate sex-specificity in the *COL11A1* and *MMP14* loci, we performed stratification-by-sex analysis in the discovery study (USA_TX). Association for the imputed allele dosages in rs3753841 and rs37538 were computed separately for females (1,157 cases and 7138 controls) using logistic regression with ten principal components as covariates in Mach2dat⁸⁴.

RNAseq of human tissues

RNAseq was performed as previously described⁹⁴. Read counting and transcript quantification were performed using HTSeq⁹⁵. Finally, reads were normalized using DESeq2 tools⁹⁶ and TPM values were generated using the Kalisto pipeline⁹⁷.

Animal studies

All mouse and rat work was conducted per IACUC approved protocols and in accordance with AALAC and NIH guidelines.

Generation of *Pax1* knockout mice

Mouse work was approved by the UCSF IACUC, protocol number AN181381. Two gRNAs were designed to target the 5' and 3' ends of *Pax1* gene (gRNA sequence shown in **Figure 5-figure supplement 5a**) using the gRNA design tool on the Integrated DNA Technologies (IDT, Newark, NJ, USA) website and selected based on low off-target and high on-target scores. The knockout allele was generated using *i*-GONAD⁹⁸ as previously described⁹⁹.

To validate proper generation of knockout, mice were analyzed by genotyping with primers shown in Appendix A Key Resources Table, Sanger sequencing of PCR-amplified DNA, and Southern blot (**Figure 5-figure supplement 5a**). For Southern blot analyses, genomic DNA were treated with NcoI (Cat #R0193, New England Biolabs, MA, USA) and fractionated by agarose gel electrophoreses. Following capillary transfer onto nylon membranes, blots were hybridized with Digoxigenin (DIG)-labeled DNA probes (corresponding to chr2:147,202,083-147,202,444; mm9) amplified by the PCR DIG Probe Synthesis Kit (Cat #11636090910, Sigma-Aldrich, MO, USA). The hybridized probe was immunodetected with antidigoxigenin Fab fragments conjugated to alkaline phosphatase (Cat # 11093274910, Sigma-Aldrich, MO, USA) and visualized with a CDP star (Cat #11685627001, Sigma-Aldrich, MO, USA) according to the manufacturer's protocol. Chemiluminescence was detected using the FluorChem E (Cat #92-14860-00, ProteinSimple, CA, USA).

***Col11a1^{fl/fl}* mice**

The *Col11a1^{fl/fl}* mouse line was kindly provided by Dr. Lou Soslowsky with permission from Dr. David Birk. All *Col11a1^{fl/fl}* and wildtype mice were handled per UTSW IACUC, protocol number 2016-101455.

Other mice

Cartilage was harvested from C57B46 wild type mice for siRNA-mediated knockdowns experiments.

Histologic methods

For thin cryostat sections, P0.5 mouse whole body was fixed in 4% paraformaldehyde for 6 hours followed by 10% sucrose for 12 hours, then transferred to 18% sucrose for 24 hours. Tissues were then embedded in optimal cutting temperature compound (OCT) and sectioned using low-profile blades on a Thermo Shandon Cryostar NX70 cryostat and all sections were lifted on APES clean microscope slides. For whole mount images, samples were treated similarly with the addition of 2% polyvinylpyrrolidone (PVP) during the cryoprotection step and frozen in 8% gelatin (porcine) in the presence of 20% sucrose and 2% PVP. Samples were sectioned at a thickness of 10 μ m. Slides were stored at -80° C until ready for use. For P28 and older mice, spines were removed then fixed, decalcified, and embedded in OCT. Spines were processed by making 7 μ m thick lateral cuts the length of the spine.

Collagen α 1(XI) was detected by immunohistochemistry staining using affinity-purified antisera against peptide (C) YGTMEPYQTETPRR-amide (Genescript, NJ, USA) as described⁴⁰ and secondary horseradish peroxidase (HRP) conjugated affinity purified secondary antibody (Cat # AP187P, Millipore-Sigma Aldrich, MO, USA). Briefly, frozen sections were equilibrated to room temperature for 1 hour, then fixed with 4% paraformaldehyde in PBS at 4 degrees Celsius for 20 minutes. Slides were washed, treated with 3% H_2O_2 in methanol for 10 minutes to block endogenous peroxidase, washed, and transferred to PBS with 0.05% TWEEN 20 (Cat #P3563-10PAK, Sigma-Aldrich, MO, USA) pH 7.4. Slides were blocked with 0.5 % goat serum in PBS mix with 0.2% Triton 100 (Cat #T8787, Sigma-Aldrich, MO, USA) at room temperature for 1.5 hours. The primary collagen α 1(XI) affinity purified antibody was applied at 0.40 mg/ml and slides were incubated overnight at 4°C. Afterward slides were washed in PBS Tween 20 for three times and treated with goat anti rabbit-HRP for 1.5 hours, then washed three times in PBS Tween 20. After applying 3,3'-diaminobenzidine (DAB) solution, slides were washed and counterstained with Mayer's hematoxylin (Cat #MHS80, Sigma-Aldrich, MO, USA), washed, dehydrated, and mounted.

For collagen α 1(XI) and Pax1 IF studies, P0.5 mice, P28 spine and ribs sections were fixed in 4% paraformaldehyde (PFA) for 20 minutes then washed with PBS + 0.1% Triton three times, before incubation with 10% normal goat serum in PBS + 0.1% Triton for 30 min to block the background.

Slides were incubated with goat anti-mouse collagen $\alpha 1(XI)$ antibody at 1:500 dilution and mouse anti-rat Pax1 (Cat #MABE1115M, Sigma-Aldrich, MO, USA), in PBS + 0.1% Triton + 1% normal goat serum at 4 °C overnight. Secondary antibodies used were 1:5000 anti-rat Alexa488 and anti mouse Alexa594 conjugated antibodies (Cat #A32740 Invitrogen, CA, USA). The sections were mounted using ProLong Gold with DAPI (Cat #S36964 Invitrogen, CA, USA) for imaging as described.¹⁰⁰

#S36964 Invitrogen, CA, USA)). All images were taken with Carl Zeiss Axio Imager.M2 fluorescence microscope (Zeiss, Oberkochen, DE).

Rib cartilage and IVD cell culture

Mouse rib cartilage cells were isolated from the rib cage and sternum of P0.5 mice. Rat IVD was removed intact from 1-month female rats and immediately separated into NP, AF, and CEP isolates. Subsequently, tissues were incubated and shaken with 2 mg/ml Pronase solution (Cat #10165921001 Sigma-Aldrich, Inc., St. Louis, MO, USA) for 1 hour, followed by 1.5 hours digestion with 3 mg/ml Collagenase D solution. (Cat #11088882001 Sigma-Aldrich, Inc., St. Louis, MO, USA), then 5 hours digestion with 0.5 mg/ml Collagenase D solution before 3 times PBS wash. Filtered, dissociated cells were seeded in Dulbecco's modified Eagle's medium (DMEM; Cat #MT15017CV Thermo Fisher Scientific, MA, USA) containing 10% fetal bovine serum (FBS), 100 μ g/ml streptomycin and 100 IU/ml penicillin. Remaining cartilage tissues underwent further digestion in 86 U/ml type 1 collagenase (Cat # SCR103 Sigma-Aldrich, Inc., St. Louis, MO, USA) overnight and cells collected.

SV40 immortalization of primary chondrocytes

Col11a1^{f/f} mouse rib cartilage cells were isolated from the rib cage and sternum of P0.5 mice. The cells were transduced with pRRLsin-sv40 T antigen-IRES-mCherry lenti-virus.¹⁰¹ for 48 hours, then sorted for mCherry-positive cells by flow cytometry. mCherry-positive cells were then infected with plv-eGFP, plv-eGFP-COL11A1-HA, plveGFP-COL11A1^{P1335L}-HA constructs. After expansion, GFP-positive cells were sorted by flow cytometry.

Adenovirus treatment

SV40 induced *Col11a1*^{f/f} mouse rib cartilage cells seeded on 6 wells plate and grown to 50% confluency. Afterward, cells were treated with 2ul Ad5-CMV-cre adenovirus (titer 1.8×10^{11} pfu/ml) and Ad5-CMV-eGFP (titer 1.65×10^{10} pfu/ml) as control. Both virus strains were from the Gene Vector Core facility, Baylor College of Medicine. After 48-hours the cells were harvested for mRNA and protein lysate.

RNA-seq and qRT-PCR

For *Pax1* knockout studies, total RNA was collected from E12.5 tails using TRIzol (Cat #15596026, Thermo Fisher Scientific, MA, USA) and converted to cDNA using ReverTra Ace qPCR-RT master mix with genomic DNA (gDNA) remover (Cat #FSQ-301, Toyobo, Osaka, Japan). Sequencing was done using an Illumina Novaseq platform and the data were analyzed using Partek Flow (Version 10.0) and Gene ontology.¹⁰² qPCR was performed using SsoFast EvaGreen supermix (Cat #1725205, Bio-Rad, CA, USA). Primer sequences used for qPCR are shown in Appendix A, Key Resources table.

To quantify the expression level of *Col11a1*, *Mmp3*, and marker genes in IVD compartments and rib cartilage, cultured cells were collected in RNeasy (Qiagen, Inc.) for RNA purification. Taqman Reverse Transcription Kit (Cat #4387406 Thermo Fisher Scientific, MA, USA) was used to reverse transcribe mRNA into cDNA. Following this, RT-qPCR was performed using a Power SYBR Green PCR Master Mix Kit (Cat #1725271, Bio-Rad, CA, USA). The primer sequences for the genes used in this study are listed in the Appendix A Key Resources Table. Gene expression was calculated using the $\Delta\Delta CT$ method after normalizing to GAPDH.

siRNA knockdown

Mouse rib cartilage cells seeded in 6 well-plates were 60-80% confluent at transfection. Lipofectamine RNAiMAX reagent (Cat #13778030 Thermo Fisher, Waltham, MA, USA) was diluted (9ul in 500ul) in Opti-MEM Medium (Cat # 31985070 Thermo Fisher MA, USA). 30 pmol siRNA was diluted in 500ul Opti-MEM medium, then added to diluted Lipofectamine RNAiMAX Reagent. siRNA-lipid complex was added to cells after 5-minute incubation at room temperature. Cells were harvested after 72 hours.

Western blotting

For Mmp3 western blotting, a total of 30 µg protein mixed with SDS-PAGE buffer was loaded on 12% SDS-polyacrylamide gel for electrophoresis. For collagen α1(XI) western blotting, 50 µg protein mixed with SDS-PAGE buffer was loaded on 4-20% SDS-polyacrylamide gel. The separated proteins were then transferred to nitrocellulose membranes (Cat #77010 Thermo Fisher Waltham, MA, USA) at 100 volts for 2-3 hours. The membrane was first incubated with blocking buffer containing 5% defatted milk powder, and then exposed to 0.1 mg/mL mouse anti-rabbit Mmp3 (Cat #ab214794 Abcam, Cambridge, MA, USA) or anti-rabbit Col11a1 (Cat #PA5-38888 Thermo Fisher Waltham, MA, USA) overnight. The samples were then washed thoroughly with TBS buffer, followed by incubation with HRP-labeled antirabbit IgG secondary antibodies 1:5000, (Cat #32460 Thermo Fisher Waltham, MA, USA) overnight. The membranes were then washed with TBS buffer. GAPDH was detected by a rabbit anti-mouse antibody (Cat #14C10 Cell Signaling, MA, USA) and used as the internal loading control.

Data availability

Summary data for GWAS3 is deposited in the NHGRI-EBI GWAS catalog (accession #GCST90179478). Otherwise, data generated or analyzed in this study are included in the manuscript.

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Adolescent Idiopathic Scoliosis 1000 Exomes Study (phs001677.v1.p1) which included investigators at Washington University in St Louis (M. Dobbs), Shriners Hospital for Children, St Louis (M. Dobbs), University of Colorado (N. Miller), University of Iowa (S. Weinstein and J. Morcuende), University of Wisconsin (P. Giampietro), and Hospital for Special Surgery (C. Raggio). Sequencing of these samples was funded by NIH NIAMS 1R01AR067715 (M. Dobbs and CG). Use of the dbGAP dataset phs001039.v1.p1 is gratefully acknowledged. All contributing sites and additional funding information for dbGAP dataset phs001039.v1.p1 are acknowledged in this publication: Fritsche et al. (2016) Nature Genetics 48 134–143, (doi:10.1038/ng.3448); The International AMD Genomics consortium's web page is: http://eaglep.case.edu/iamdgc_web/, and additional information is available on: <http://csg.sph.umich.edu/abecasis/public/amd015/>. The authors acknowledge the Texas Advanced Computing Center (TACC) at The University of Texas at Austin for providing computing resources that have contributed to the research results reported within this paper. URL: <http://www.tacc.utexas.edu>

Author contributions

H.Y., A.M.K., A.U., X-Y. L, N.O, Y.K., E.E., Y.F., L.A., Y.H.K., R.C., R.S, Y.Z., J.P. performed genetic, bioinformatic, or proteomic analyses or molecular biology experiments. N.V.G., B.M.E., J.P.Y.C., and J.A.H. provided technical or clinical advice. C.T., Y-Q.S., C.A.G., P.G., and S.I. supervised genetic experiments and provided summary data. J.J.R., N.A., and C.A.W. supervised the work. C.A.W conceived the study and wrote the manuscript. All co-authors reviewed and approved the final manuscript.

Competing interests

The authors declare no competing interests.

Appendix A. Key Resources Table

siRNA and Primer Sequences

Appendix B. Clinical groups

Texas Scottish Rite Hospital for Children Clinical Group (TSRHCCG)

Lori A. Karol¹, Karl E. Rathjen¹, Daniel J. Sucato¹, John G. Birch¹, Charles E. Johnston III¹, Benjamin S. Richards¹, Brandon Ramo¹, Amy L. McIntosh¹, John A. Herring¹, Todd A. Milbrandt², Vishwas R. Talwakar³, Henry J. Iwinski^{4,2}, Ryan D. Muchow³, J. Channing Tassone⁴, X.-C. Liu⁴, Richard Shindell⁵, William Schrader⁶, Craig Eberson⁷, Anthony Lapinsky⁸, Randall Loder⁹ and Joseph Davey¹⁰

1. Department of Orthopaedic Surgery, Texas Scottish Rite Hospital for Children, Dallas, Texas, USA.
2. Department of Orthopaedic Surgery, Mayo Clinic, Rochester, Minnesota, USA
3. Department of Orthopaedic Surgery, Shriners Hospitals for Children, Lexington, Kentucky, USA.
4. Department of Orthopaedic Surgery, Children's Hospital of Wisconsin, Milwaukee, Wisconsin, USA.
5. OrthoArizona, Phoenix, Arizona, USA.
6. Departments of Orthopedics, Sports Medicine, and Surgical Services, Akron Children's Hospital, Akron, Ohio, USA.
7. Pediatric Orthopaedics and Scoliosis, Hasbro Children's Hospital, Providence, Rhode Island, USA.
8. University of Massachusetts Memorial Medical Center, Worcester, Massachusetts, USA.
9. Indiana University-Purdue University Indianapolis, Indianapolis, Indiana, USA.
10. University of Oklahoma Health Sciences Center, Oklahoma City, Oklahoma, USA.

Japan Scoliosis Clinical Research Group (JSCRG)

Kota Watanabe¹, Nao Otomo^{1, 2}, Kazuki Takeda^{1, 2}, Yoshiro Yonezawa^{1, 2}, Yoji Ogura^{1, 2}, Yohei Takahashi^{1, 2}, Noriaki Kawakami³, Taichi Tsuji⁴, Koki Uno⁵, Teppei Suzuki⁵, Manabu Ito⁶, Shohei Minami⁷, Toshiaki Kotani⁷, Tsuyoshi Sakuma⁷, Haruhisa Yanagida⁸, Hiroshi Taneichi⁹, Satoshi Inami⁹, Ikuho Yonezawa¹⁰, Hideki Sudo¹¹, Kazuhiro Chiba¹², Naobumi Hosogane¹², Kotaro Nishida¹³, Kenichiro Kakutani¹³, Tsutomu Akazawa¹⁴, Takashi Kaito¹⁵, Kei Watanabe¹⁶, Katsumi Harimaya¹⁷, Yuki Taniguchi¹⁸, Hideki Shigemats¹⁹, Satoru Demura²⁰, Takahiro Iida²¹, Ryo Sugawara²², Katsuki Kono²³, Masahiko Takahata²⁴, Norimasa Iwasaki²⁴, Eijiro Okada¹, Nobuyuki Fujita¹, Mitsuru Yagi¹, Masaya Nakamura¹, Morio Matsumoto¹

1. Department of Orthopaedic Surgery, Keio University School of Medicine, Tokyo, Japan
2. Laboratory of Bone and Joint Diseases, Center for Integrative Medical Sciences, RIKEN, Tokyo, Japan

3. Department of Orthopaedic Surgery, Meijo Hospital, Nagoya, Japan
4. Department of Orthopaedic Surgery, Toyota Kosei Hospital, Nagoya, Japan
5. Department of Orthopaedic Surgery, National Hospital Organization, Kobe Medical Center, Kobe, Japan
6. Department of Orthopaedic Surgery, National Hospital Organization, Hokkaido Medical Center, Sapporo, Japan
7. Department of Orthopaedic Surgery, Seirei Sakura Citizen Hospital, Sakura, Japan
8. Department of Orthopaedic Surgery, Fukuoka Children's Hospital, Fukuoka, Japan
9. Department of Orthopaedic Surgery, Dokkyo Medical University School of Medicine, Mibu, Japan
10. Department of Orthopaedic Surgery, Juntendo University School of Medicine, Tokyo, Japan
11. Department of Advanced Medicine for Spine and Spinal Cord Disorders, Hokkaido University Graduate School of Medicine, Sapporo, Japan
12. Department of Orthopaedic Surgery, National Defense Medical College, Tokorozawa, Japan
13. Department of Orthopaedic Surgery, Kobe University Graduate School of Medicine, Kobe, Japan
14. Department of Orthopaedic Surgery, St. Marianna University School of Medicine, Kawasaki, Japan
15. Department of Orthopaedic Surgery, Osaka University Graduate School of Medicine, Suita, Japan
16. Department of Orthopaedic Surgery, Niigata University Hospital, Niigata, Japan
17. Department of Orthopaedic Surgery, Graduate School of Medical Sciences, Kyushu University Beppu Hospital, Fukuoka, Japan
18. Department of Orthopaedic Surgery, Faculty of Medicine, The University of Tokyo, Tokyo, Japan
19. Department of Orthopaedic Surgery, Nara Medical University, Nara, Japan
20. Department of Orthopaedic Surgery, Kanazawa University School of Medicine, Kanazawa, Japan
21. Department of Orthopaedic Surgery, Dokkyo Medical University Koshigaya Hospital, Koshigaya, Japan
22. Department of Orthopaedic Surgery, Jichi Medical University, Shimotsuke, Japan
23. Department of Orthopaedic Surgery, Kono Orthopaedic Clinic, Tokyo, Japan
24. Department of Orthopaedic Surgery, Hokkaido University, Sapporo, Japan

Scoliosis and Genetics in Scandinavia (ScoliGeneS) study group

Tian Cheng¹, Juha Kere², Aina Danielsson³, Kristina Åkesson⁴, Ane Simony⁵, Mikkel Andersen⁵, Steen Bach Christensen⁵, Maria Wikzén⁶, Luigi Belcastro⁶

1. Department of Clinical Sciences and Technology, Karolinska Institutet, Huddinge, Sweden
2. Department of Biosciences and Nutrition, Karolinska Institutet, Huddinge, Sweden
3. Department of Orthopedics, Sahlgrenska University Hospital, Gothenburg, Sweden
4. Department of Orthopedics and Clinical Sciences, Lund University, Skane University Hospital, Malmö, Sweden
5. Sector for Spine Surgery and Research, Middelfart Hospital, Middelfart, Denmark
6. Department of Reconstructive Orthopaedics, Karolinska University Hospital, Stockholm, Sweden.

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Author information

Hao Yu

Center for Pediatric Bone Biology and Translational Research, Scottish Rite for Children, Dallas, TX, USA

Anas M. Khanshour

Center for Pediatric Bone Biology and Translational Research, Scottish Rite for Children, Dallas, TX, USA

Aki Ushiki

Department of Bioengineering and Therapeutic Sciences, University of California San Francisco, San Francisco, CA, USA, Institute for Human Genetics, University of California San Francisco, San Francisco, CA, USA

Nao Otomo

Laboratory of Bone and Joint Diseases, RIKEN Center for Integrative Medical Sciences, Tokyo, JP

Yoshinao Koike

Laboratory of Bone and Joint Diseases, RIKEN Center for Integrative Medical Sciences, Tokyo, JP, Laboratory for Statistical and Translational Genetics, RIKEN Center for Integrative Medical Sciences, Yokohama, JP

Elisabet Einarsdottir

Science for Life Laboratory, Department of Gene Technology, KTH-Royal Institute of Technology, Solna, SE

Yanhui Fan

School of Biomedical Sciences, The University of Hong Kong, Hong Kong SAR, CN

Lilian Antunes

Department of Neurology, Washington University in St. Louis, St. Louis, MO, USA

Yared H. Kidane

Center for Pediatric Bone Biology and Translational Research, Scottish Rite for Children, Dallas, TX, USA

Reuel Cornelia

Center for Pediatric Bone Biology and Translational Research, Scottish Rite for Children, Dallas, TX, USA

Rory Sheng

Department of Bioengineering and Therapeutic Sciences, University of California San Francisco, San Francisco, CA, USA, Institute for Human Genetics, University of California San Francisco, San Francisco, CA, USA

Yichi Zhang

Department of Bioengineering and Therapeutic Sciences, University of California San Francisco, San Francisco, CA, USA, Institute for Human Genetics, University of California San Francisco, San Francisco, CA, USA, School of Pharmaceutical Sciences, Tsinghua University, Beijing, CN

Jimin Pei

Department of Biophysics, University of Texas Southwestern Medical Center, Dallas, TX, USA

Nick V. Grishin

Department of Biophysics, University of Texas Southwestern Medical Center, Dallas, TX, USA

Bret M. Evers

Department of Pathology, University of Texas Southwestern Medical Center, Dallas, TX, USA,
Department of Ophthalmology, University of Texas Southwestern Medical Center, Dallas, TX, USA

Jason Pui Yin Cheung

Department of Orthopaedics and Traumatology LKS Faculty of Medicine, The University of Hong Kong, Hong Kong SAR, CN

John A. Herring

Department of Orthopedic Surgery, Scottish Rite for Children, Dallas, TX, USA, Department of Orthopaedic Surgery, University of Texas Southwestern Medical Center, Dallas, TX, USA

Chikashi Terao

Laboratory for Statistical and Translational Genetics, RIKEN Center for Integrative Medical Sciences, Yokohama, JP

You-Qiang Song

School of Biomedical Sciences, The University of Hong Kong, Hong Kong SAR, CN

Christina A. Gurnett

Department of Neurology, Washington University in St. Louis, St. Louis, MO, USA

Paul Gerdhem

Department of Clinical Science, Intervention & Technology (CLINTEC), Karolinska Institutet, Stockholm, Uppsala University, Uppsala, SE, Department of Surgical Sciences, Uppsala University and, Department of Orthopaedics and Hand Surgery, Uppsala University Hospital, Uppsala, SE

Shiro Ikegawa

Laboratory of Bone and Joint Diseases, RIKEN Center for Integrative Medical Sciences, Tokyo, JP

Jonathan J. Rios

Center for Pediatric Bone Biology and Translational Research, Scottish Rite for Children, Dallas, TX, USA, Department of Orthopaedic Surgery, University of Texas Southwestern Medical Center, Dallas, TX, USA, Eugene McDermott Center for Human Growth and Development, University of Texas Southwestern Medical Center, Dallas, TX, USA, Department of Pediatrics, University of Texas Southwestern Medical Center, Dallas, TX, USA

Nadav Ahituv

Department of Bioengineering and Therapeutic Sciences, University of California San Francisco, San Francisco, CA, USA, Institute for Human Genetics, University of California San Francisco, San Francisco, CA, USA

Carol A. Wise

Center for Pediatric Bone Biology and Translational Research, Scottish Rite for Children, Dallas, TX, USA, Department of Orthopaedic Surgery, University of Texas Southwestern Medical Center, Dallas, TX, USA, Eugene McDermott Center for Human Growth and Development, University of Texas Southwestern Medical Center, Dallas, TX, USA, Department of Pediatrics, University of Texas Southwestern Medical Center, Dallas, TX, USA

For correspondence: carol.wise@tsrh.org

ORCID iD: [0000-0002-6790-2194](https://orcid.org/0000-0002-6790-2194)

Editors

Reviewing Editor

Michel Bagnat

Duke University, United States of America

Senior Editor

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University of Barcelona, Spain

Joint Public Review:

Summary: This study follows up on previous work showing a female-specific enhancer region of PAX1 is associated with adolescent idiopathic scoliosis (AIS). This new analysis combines human GWAS analysis from multiple countries to identify a new AIS-associated coding variant in the COL11A1 gene. Two nonsynonymous variants were found to be significantly associated with AIS: MMP14 p.Asp273Asn and COL11A1 p.Pro1335Leu, the latter of which had the more robust association and remained significant when females were tested independent of males. Using a Pax1 knockout mouse they go on to find that PAX1 and Collagen XI protein are expressed in the intervertebral discs (IVDs) and robustly in the growth plate, showing that COL11A1 expression is reduced in Pax1 mutant growth plate. Moreover, other AIS-associated genes, Gpr126 and Sox6, were also reduced in Pax1 mutant mice, suggesting a common pathway is involved in AIS. The proposed implication of a Pax1-Col11a1-Mmp3 signaling axis modulated by estrogen signaling suggests a potential mechanism by which young women have more severe scoliosis than young men, as is observed in humans.

Strengths: This work integrates a large cohort of human genetic data from AIS patients and controls from diverse ethnic backgrounds, across the globe. This work attempts to functionally test their findings in vivo and by use of cell culture. The authors propose an interesting model which warrants in depth investigation.

Weaknesses: There are concerns regarding the candidacy of COL11A1 p.Pro1335Leu that need to be addressed and clarified. Many of the main functional work was done in cell culture and not in vivo. Moreover, the evidence linking COL11A1 p.Pro1153Leu to AIS is indirect, making unclear whether impaired COL11A1 function can cause scoliosis in the mouse model, thus diminishing the strength of the conclusions regarding the proposed pathogenicity of COL11A1 p.Pro1335Leu.

Author Response

We thank the reviewers for truly valuable advice and comments. We have made multiple corrections and revisions to the original pre-print accordingly. Here we address 2 major points.

1. Regarding the genetic association of the common COL11A1 variant rs3753841 (p.(Pro1335Leu)), we do not propose that it is the sole risk variant contributing to the association signal we detected and have clarified this in the manuscript. We concluded that it was worthy of functional testing for reasons described here. Although there were several common variants in the discovery GWAS within and around COL11A1, none were significantly associated with AIS and none were in linkage disequilibrium ($R^2 > 0.6$) with the top SNP rs3753841. We next reviewed rare ($MAF \leq 0.01$) coding variants within the COL11A1 LD region of the associated SNP (rs3753841) in 625 available exomes representing 46% of the 1,358 cases from the discovery cohort. The LD block was defined using Haploview based on the 1KG_CEU population. Within the ~41 KB LD region (chr1:103365089- 103406616, GRCh37) we found three rare missense mutations in 6 unrelated individuals, Author response table 1. Two of them (NM_080629.2: c.G4093A:p.A1365T; NM_080629.2:c.G3394A:p.G1132S), from two individuals, are predicted to be deleterious based on CADD and GERP scores and are plausible AIS risk candidates. At this rate we could expect to find only 4-5 individuals with linked rare coding variants in the total cohort of 1,358 which collectively are unlikely to explain the overall association signal we detected. Of course, there also could be deep intronic variants contributing to the association that we would not detect by our methods. However, given this scenario, the relatively high predicted deleteriousness of rs3753841 (CADD= 25.7; GERP=5.75), and its occurrence in a Gly-X-Y triplet repeat, we hypothesized that this variant itself could be a risk allele worthy of further investigation.

Author response table 1.

VarID_GRCh37	RsID	Mutation	gnomAD	CADD	GERP	#Cases
1:103377744:C:T	rs151249006	NM_080629.2:c.G4093A:p.A1365T	6.37E-05	25.8	5.42	1
1:103381192:C:A	rs150669855	NM_080629.2:c.G3847T:p.V1283L	0.0007	0.001	-10.9	4
1:103405909:C:T	rs370589018	NM_080629.2:c.G3394A:p.G1132S	6.37E-05	25.2	5.46	1

We also appreciate the reviewer's suggestion to perform a rare variant burden analysis of COL11A1. We conducted pilot gene-based analysis in 4534 European ancestry exomes including 797 of our own AIS cases and 3737 controls and tested the burden of rare variants in COL11A1. SKATO P value was not significant (COL11A1_P=0.18) but this could be due to lack of power and/or background from rare benign variants that could be screened out using the functional testing we have developed.

1. Regarding functional testing, by knockdown/knockout cell culture experiments, we showed for the first time that Col11a1 negatively regulates Mmp3 expression in cartilage chondrocytes, an AIS-relevant tissue. We then tested the effect of overexpressing the human wt or variant COL11A1 by lentiviral transduction in SV40-transformed chondrocyte cultures. We deleted endogenous mouse Col11a1 by Cre recombination to remove the background of its strong suppressive effects on Mmp3 expression. We acknowledge that Col11a1 missense mutations could confer gain of function or dominant negative effects that would not be revealed in this assay. However as indicated in our original manuscript we have noted that spinal deformity

is described in the cho/cho mouse, a Col11a1 loss of function mutant. We also note the recent publication by Rebello et al. showing that missense mutations in Col11a2 associated with congenital scoliosis fail to rescue a vertebral malformation phenotype in a zebrafish col11a2 KO line. Although the connection between AIS and vertebral malformations is not altogether clear, we surmise that loss of the components of collagen type XI disrupt spinal development. *in vivo* experiments in vertebrate model systems are needed to fully establish the consequences and genetic mechanisms by which COL11A1 variants contribute to an AIS phenotype.